MODULE 36: SPLUNK TROUBLESHOOTING

36.0 The master workflow

Always follow the data path from **source to search**. Find where it stops.

```
1. Is the SOURCE producing data?          (file growing? app logging?)
2. Is the INPUT configured + readable?    (inputs.conf, permissions)
3. Is the FORWARDER connected?            (outputs.conf, port 9997)
4. Did the INDEXER RECEIVE it?            (_internal metrics, tcpin)
5. Is it in the RIGHT INDEX?              (index exists, spelling)
6. Is the TIME right?                     (timestamp, time range)
7. Can I SEARCH it?                       (role permissions, index allowed)
8. Are FIELDS extracted?                  (sourcetype, props, permissions)
```

Top tools: `splunk btool` , `index=_internal` , `splunk list forward-server` , the Job Inspector, `ss -tlnp` , `ls -l` .

36.1 “No events found” (most important)

SYMPTOM: search returns “No results found.”
POSSIBLE CAUSES: wrong time range; wrong index name; role can’t search that index; data never arrived; source isn’t producing; typo or wrong case in a field value.
COMMANDS/SPL:

```spl
| tstats count WHERE index=* BY index
| metadata type=sourcetypes index=*
index=* sourcetype=<mine> earliest=0
```

CONFIG CHECK: `splunk btool inputs list --debug` ; Settings → Indexes; role’s allowed indexes.
FIX: widen the time to All time, correct the index or sourcetype, grant the role access, or fix the input.
VERIFY: the search returns events, and `_time` looks right.

36.2 “Index does not exist”

SYMPTOM: the input or search refers to an index and the data is missing; `splunkd.log` has a warning.
CAUSES: the index was never created on the indexer; typo; it was created on a search head only; or a forwarder points to an index that exists only on another instance. **SPL:**

```spl
index=_internal source=*splunkd.log* "unconfigured/disabled/deleted index"
| rest /services/data/indexes splunk_server=local | table title disabled
```

FIX: create the index **on the indexers** (`indexes.conf` or UI) and restart if needed.
VERIFY: new events appear. Events already dropped are lost, so re-ingest if possible.

36.3 "Data is in the wrong index"

CAUSES: missing `index =` in the input (defaults to `main`); another stanza overriding it; a transform routing it.
CHECK: `splunk btool inputs list monitor --debug`

```spl
index=* source="/var/log/auth.log" | stats count by index
```

FIX: set `index` in the correct stanza, restart, and note that **old events stay where they are** (copying or re-indexing is needed to move them).

36.4 "Wrong sourcetype"

SYMPTOM: events broken wrongly, odd fields, no timestamps.
CAUSES: automatic sourcetype guessing; no explicit sourcetype; a rule in `props.conf` overriding.

```spl
index=* source="<path>" | stats count by sourcetype
```

CHECK: `btool inputs list --debug` and `btool props list <sourcetype> --debug`.
FIX: set `sourcetype =` explicitly in `inputs.conf`, and restart. Only new events are affected.

36.5 "Fields are missing"

CAUSES: wrong sourcetype; no extraction defined; the regex doesn't match; extraction is private/not shared; the field exists only in some events; the add-on is not installed on the search head.
CHECK:

```spl
index=<idx> sourcetype=<st> | fieldsummary maxvals=5 | table field count distinct_count
```

Check the search mode (**Verbose** shows all fields, while "Fast" mode hides some). Settings → Fields → Field extractions → permissions.
FIX: correct the sourcetype, install the add-on on the **search head**, add an extraction (`EXTRACT`in props), and share it.

36.6 "Timestamp is incorrect"

SYMPTOM: events appear hours off, in the future, or at index time.
CAUSES: time zone mismatch; wrong `TIME_FORMAT` or `TIME_PREFIX`; Splunk found a different date in the text; no timestamp in the event (uses time received).

```spl
index=<idx> | eval lag=_indextime-_time | stats avg(lag) max(lag) by sourcetype
index=_internal component=DateParserVerbose
```

CONFIG:

```ini
[my_app_logs]
TIME_PREFIX = ^\[
TIME_FORMAT = %Y-%m-%d %H:%M:%S
MAX_TIMESTAMP_LOOKAHEAD = 19
TZ = Asia/Kolkata
```

(use the time zone of the log's source; `Asia/Kolkata` is only an example)
FIX: place on the **indexer/HF (parsing tier)**, restart. **Only new events** are corrected.
VERIFY: `lag` is near zero.

36.7 "Forwarder is not sending"

CAUSES: UF stopped; `outputs.conf` wrong; receiver closed; firewall; UF can't read the file; no new data in the file.

On the forwarder (Linux):

```bash
sudo /opt/splunkforwarder/bin/splunk status
sudo /opt/splunkforwarder/bin/splunk list forward-server
sudo /opt/splunkforwarder/bin/splunk btool outputs list --debug
tail -50 /opt/splunkforwarder/var/log/splunk/splunkd.log
```

`list forward-server` shows **Active** vs **Configured but inactive** forwards.

On the indexer:

```spl
index=_internal source=*metrics.log* group=tcpin_connections
| stats latest(_time) AS last_seen by hostname, sourceIp
```

FIX: start the UF, correct `server=` in `outputs.conf` , restart, and open the network path. **VERIFY:** the host appears in the search above and events arrive.

36.8 "Receiver is not accepting" and 36.9 "Port 9997 problem"

SYMPTOM: UF log: "Connection refused," "Cannot connect to…," or "Connected but no data."
CAUSES: receiving port not enabled on the indexer; the port is enabled but a **firewall blocks it** (the port can be "open" in a sense but not for your source); SSL mismatch; the indexer is down or busy (queues blocked); wrong IP/DNS.
COMMANDS:

```bash
ss -tlnp | grep 9997                    # on indexer: is Splunk listening?
nc -zv <indexer-ip> 9997                # on forwarder: is the port reachable?
sudo ufw status                         # Ubuntu firewall
sudo /opt/splunk/bin/splunk display listen -auth admin:<pw>
```

```spl
index=_internal source=*splunkd.log* component=TcpInputProc
index=_internal source=*metrics.log* group=queue blocked=true
```

FIX: `splunk enable listen 9997`, open the firewall, correct IP/port, fix SSL settings, and clear indexer queue problems.
Key idea: "port reachable" ≠ "data flowing." Check the UF's `splunkd.log`, the indexer's `tcpin_connections`, and queues.

36.10 "HEC problem"

CAUSES: HEC not enabled globally; token disabled; **index not allowed for the token**; wrong token or header; wrong endpoint or JSON; HTTPS vs HTTP mismatch (HEC uses SSL by default).
TEST:

```bash
curl -k https://localhost:8088/services/collector/health
curl -k https://localhost:8088/services/collector/event \
  -H "Authorization: Splunk <token>" \
  -d '{"event":"hello","index":"main","sourcetype":"hec_test"}'
```

- `-k` skips certificate checking (lab only). `-H` sets the header.
- Success returns `{"text":"Success","code":0}`. Error responses name the problem (bad token, disabled, invalid index, bad JSON).
  FIX: Settings → Data inputs → HTTP Event Collector → enable Global Settings, check the token's **allowed indexes** and default index, fix the URL/token.

36.11 "Search is slow"

CAUSES: broad base search (`index=*`, All time); leading wildcards; `join` / `transaction` / `map`; many unneeded fields; high-cardinality `by`; too many concurrent searches.
TOOLS: Job → Inspect Job (execution costs: which phase took the time), and search mode **Fast**.
FIX:

- Narrow `index`, `sourcetype`, and time first. `fields` early to drop unneeded fields.
- Replace `join` / `transaction` with `stats`.
- Use `tstats` /accelerated data models for counts.
- Use summary indexing/scheduled searches for heavy dashboards. **VERIFY:** compare the job inspector before and after.

36.12 "Lookup not working"

CAUSES: file not uploaded or definition missing; not shared; field name mismatch; case/whitespace; automatic lookup is bound to the wrong sourcetype. **CHECK:**

```spl
| inputlookup users_lookup
```

Then compare field values: `| eval check=len(user)` to spot hidden spaces.
FIX: share permissions, add `AS`, normalize with `lower()` / `trim()`, or set `case_sensitive_match=false` on the definition.

36.13"Regex not extracting"

CHECK: test with `| head 20 | rex ... | table _raw field`; escape special characters; make variable parts optional; check for a missing `?<name>`; check `field=`.

```spl
... | rex "user=(?<user>\w+)" | where isnull(user)
```

This shows the events your regex fails on.
FIX: adjust the regex for the variants, then make it permanent via props or the Field Extractor and share it.

36.14 "Props/transforms not working"

CAUSES: wrong **tier** (parsing settings on a UF, or on a search head when the data goes through a

```bash
sudo /opt/splunk/bin/splunk btool props list <sourcetype> --debug
sudo /opt/splunk/bin/splunk btool transforms list <transform_name> --debug
sudo /opt/splunk/bin/splunk btool check
```

different indexer); the stanza name doesn't match the sourcetype/source/host; no restart; **the transform name in props doesn't match** `transforms.conf`; regex mismatch; wrong `DEST_KEY`.
CHECK:

FIX: put the config on the first full Splunk instance in the path (indexer or HF), restart, and test with **new** events.
Remember: index-time changes don't change old data.

36.15 "Configuration change not taking effect"

Checklist:

1. Did you edit `local/`, not `default/`? (btool shows the winner)
2. Right **app** and **instance** (UF vs HF vs indexer vs search head)?
3. Did you **restart** (index-time) or refresh (search-time)?
4. Typo in stanza or attribute? (`btool check`)
5. Overridden by a higher-precedence file? (`btool --debug`)
6. Testing with **old** events? (index-time changes affect only new ones)

36.16 "Permission denied"

Two different types, and you must separate them:

- **OS-level:** Splunk can't read a file.

```bash
ls -l /var/log/auth.log
sudo -u splunk head -1 /var/log/auth.log
```

Fix with group membership (for example `adm`), file ACLs (`setfacl`), or log permissions, and restart.

```spl
index=_internal source=*splunkd.log* "Permission denied"
```

- **Splunk-level:** a user can't search an index or open an object. Check **Settings** → **Roles** (allowed indexes, capabilities) and the object's **Permissions** (Private/App/All apps).

36.17 "Dashboard panel not loading"

CAUSES: search error or timeout; token not set; wrong token name; invalid XML; the underlying report/lookup is private; the viewer's role lacks index access; heavy search.
CHECK: click the panel's **Open in Search**, and run it with the token values replaced. Check the browser console and **Edit** → **Source** for XML errors.
FIX: set token defaults, share the objects, optimize or accelerate the search.

36.18 "Alert not triggering"

See **Module 28.10**. Quick path: run the search manually with the same time range → check the schedule → `index=_internal sourcetype=scheduler savedsearch_name="..."` for `status` → trigger condition → throttle → action settings (email).

36.19 Troubleshooting Splunk itself: key logs

File (`/opt/splunk/var/log/splunk/`)	Use
`splunkd.log`	Main log (errors and warnings)
`metrics.log`	Throughput, queues, connections
`scheduler.log`	Scheduled searches/alerts
`splunkd_access.log`	Splunk Web/REST requests
`license_usage.log`	License use

```spl
index=_internal source=*splunkd.log* (log_level=ERROR OR log_level=WARN) | stats count by
```

36.20 Practice scenarios (troubleshoot before reading the fix)

Answer each with your **first 4 checks** (don't peek).

1. "My UF is running but Splunk has no events."
2. " `index=main` works but `index=web` gives nothing."
3. "My search returns events but `user` is missing."
4. "Events show tomorrow's date."
5. "My alert runs but no email arrives."
6. "A lookup works in search but not in the dashboard."

36.21 Interview Q&A

- **"No data from a forwarder, what do you do?"** Walk the path: source → input → forwarder status/outputs → network/port → indexer receiving and `tcpin_connections` → correct index → time range and permissions.
- **How do you find the effective config?** `splunk btool <file> list --debug` .
- **Where do you look for Splunk's own errors?** `index=_internal` (`splunkd.log` , `metrics.log` , `scheduler.log`).
- **Wrong timestamps?** Check `TIME_FORMAT` , `TIME_PREFIX` , `TZ` in props, on the parsing tier. Only new data is fixed.
- **A search is slow, what do you do?** Job inspector, narrow the scope, filter early, replace `join` / `transaction` , use `tstats` or accelerated data models.
- **Port 9997 is open but no data?** Verify the listener, firewall, `outputs.conf` , UF log, indexer `tcpin` metrics, queues, and index existence.

36.22 Revision

Follow the data path · `_internal` is your friend · `btool --debug` finds the winning config · index-time changes only affect new data · put parsing config on the right tier · permissions are both OS-level and Splunk-level.

☑ Self-Check (no peeking)

1. Report vs alert: what's the difference?
2. Why throttle an alert?
3. What does `$row.src_ip$` do?
4. How does risk-based alerting reduce noise?
5. UF running, nothing in Splunk. Name four checks in order.

Answer key for Module 36 exercises

1. Alert for 15+ failed logins in 10 minutes:

```spl
index=linux_auth sourcetype=linux_secure "Failed password" earliest=-10m
| rex "from (?<src_ip>\d{1,3}(?:\.\d{1,3}){3})"
| stats count by src_ip
| where count>=15
```

Schedule `*/10 * * * *`, time range last 10 minutes, trigger when results > 0 *for each result*, throttle 60 minutes by `src_ip`, action: Triggered Alerts or email.

2. Same user from 3+ IPs in an hour:

```spl
index=linux_auth sourcetype=linux_secure "Accepted password" earliest=-1h
| rex "for (?<user>\S+) from (?<src_ip>\d{1,3}(?:\.\d{1,3}){3})"
| stats dc(src_ip) AS ip_count, values(src_ip) AS ips by user
| where ip_count>=3
```

3. Drilldown: `<drilldown><set token="tok">$row.field$</set></drilldown>` on the first panel, and `depends="$tok$"` plus `$tok$` in the second panel's search.
4. `sudo /opt/splunk/bin/splunk btool inputs list "monitor:///var/log/auth.log" --debug`
5. Lifecycle: log is ingested, then a correlation search matches, creating a notable. L1 picks it up in Incident Review, assigns it, reviews context and drilldown, decides TP/FP, comments, and then escalates or closes with a disposition.

Self-check key: (1) A report displays results, an alert triggers an action on a condition. (2) Throttling stops repeat alerts for the same value. (3) It gives the clicked row's `src_ip` value. (4) Small signals add risk scores to an entity, and one alert fires when the total crosses a threshold. (5) UF status, `list forward-server` and `outputs.conf`, port reachability (`nc -zv`), indexer `tcpin_connections` and `_internal`, then index and time range.